

Omer Pinto

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Professional Summary

- Senior Software Engineer with **15+ years** building **backend systems**, **database internals**, and **distributed platforms**.
- Currently developing **Python** microservices and orchestration systems at BigArch while actively expanding into **AI Engineering**.
- Experimenting building **agentic workflows** using **LangGraph** / **LangChain**, alongside professional AI engineering coursework.
- On my current role, championing AI adoption inside the team: introducing practical AI-assisted development workflows, evaluating internal LLM stacks, and pushing org-level enablement.
- Strong foundation in **algorithms**, **performance tuning**, & **systems architecture** across C++, Python, Java, & cloud platforms.
- Seeking **AI Engineering roles** focused on intelligent products and **LLM-powered applications**.

Experience

BigArch

SENIOR SW ENGINEER

2024 - Present

- Lead backend development of large-scale **graph execution/orchestration platform** for HPC environments using **Python**, **MongoDB**, **RabbitMQ**, and microservices architecture.
- Architect and deploy containerized services in private-cloud environment using **Kubernetes** and **Docker**.
- Research and implement performance improvements achieving up to **60% performance gains**, transforming strained production system into stable, scalable platform.
- End-to-end ownership of large, complex features requiring deep system understanding and cross-stack design.
- Lead system-wide architectural refactors executed with **zero downtime** and full service continuity.
- **AI Initiatives:** Pioneer AI-assisted development in air-gapped environment; evaluate internal LLM infrastructure and build custom shim server for tool integration; hands-on learning of agentic frameworks (**LangGraph**, **LangChain**) applied to work MVPs.

Microsoft

SENIOR SW ENGINEER

2023 - 2024

- Engineer in the **vulnerability management** group, developing components in the endpoint security pipeline.
- Built **C# microservices** and data pipelines using Azure services (Blobs, Queues, CosmosDB, KustoDB).

Yahoo

PRINCIPAL SW ENGINEER

2022 - 2023

- Contributor to the **native ads** platform, developing **Java** services and **Hadoop/Spark** ETL pipelines.
- Partnered with research team to design and implement **ML/DL-driven ad-scoring algorithms**, resulting in smart ranking system.
- Handled codebase migration and handoff following **project cancellation**.

qSpark

SENIOR SW ENGINEER

2021 - 2022

- Engineer in the **trading infrastructure** team, delivering **ultra-low-latency C++** systems.
- Implemented **exchange protocols** (execution and market data) and handled production issues with global exchanges.
- Used advanced **modern C++ (14/17/20)** techniques, including template meta-programming, for performance-critical code.
- Designed a scalable subsystem enabling integration of 3rd-party trading algorithms.
- Conducted market-structure research (SEC/FINRA rules, protocols) to support full end-to-end trading solutions.

Toga Networks

HANDS-ON ARCHITECT

2019 - 2021

- Worked across multiple projects in the **cloud and IoT R&D** group, driving architecture and implementation.
- Performed **market and technology research** and delivered high-level designs for innovative systems.
- Designed semantic reasoning capabilities across model and query layers.
- Developed a log-analysis algorithm using **exponential backoff**, improving system performance.

SiSense

SENIOR ENGINEER, TECH LEAD & ARCHITECT, CTO OFFICE

2014 - 2018

- Hand-picked by CTO for advanced R&D team researching next-generation analytics platform technologies. Developed features on open-source database through rapid POCs and performance benchmarking.
- Architected and built cross-platform **ETL query engine** from scratch as sole developer; recruited and led team as tech lead. Optimized for low memory footprint, high CPU utilization, multi-core scaling, and distributed execution using **C++14/17** and advanced **template meta-programming**.
- Implemented core DB features and system optimizations using multi-core algorithms and cache-aware data structures. Designed thread management solution improving query engine performance by order of magnitude.
- Optimized query operations (join, group) on multi-billion record datasets. Implemented statistical and mathematical functions using algorithms from academic papers. Contributed to DB compiler internals and execution engine.

Early years - Allot Communications, ZRRO Technologies

FIRMWARE / EMBEDDED ENGINEER

2011-2014

- Design & implementation of **real-time algorithms & data structures** in a resource-restricted, **performance-critical** environment.
- Network protocols, coding of algorithms & systems in **C & C++** using extreme **memory/cpu optimizations** on **Linux** os.

Technical Skills

- **Python & Backend Development:** Production experience with microservices architecture, distributed systems (MongoDB, RabbitMQ, Kafka). RESTful APIs, async programming, performance optimization. Docker, Kubernetes for containerized deployments.
- **AI & LLM Systems:** Hands-on development with agentic frameworks (LangGraph, LangChain). Prompt engineering, multi-agent workflows, MCP protocol integration. Experience deploying LLM infrastructure in air-gapped environments. Active learning through professional coursework and personal projects.
- **Database Systems & Internals:** Deep experience implementing DB internals (query parsers, execution engines, optimizers, caching layers). Designed and built ETL engine from scratch. Working knowledge of PostgreSQL, MongoDB, ClickHouse, Druid, KustoDB. Query language DSL design.
- **Additional Languages:** Java (Spring, Maven, Hadoop/Spark), C# (Azure microservices), C++ (template metaprogramming, low-latency systems, multi-threading). Bash scripting.
- **Cloud & Distributed Systems:** Azure services (Blobs, Queues, CosmosDB, KustoDB), AWS (EC2, S3). Hadoop/Spark pipelines, distributed orchestration platforms.
- **Development Practices:** Git, CI/CD (Jenkins, ADO), Agile/Scrum, unit testing, system design, performance profiling.

Education

The Open University of Israel

Ra'anana, Israel

B.SC IN MATHEMATICS AND COMPUTER SCIENCE

2007 - 2010

- Graduated **summa cum laude** with GPA of **97/100**.
- Awarded place on University President's Excellence List (annual GPA ≥ 95) for four consecutive years.
- Achievement recognized in university educational portal [article](#).

Independent Work & Technical Writing

AI Projects, Udemy AI Learning & Technical Writing

- **LangGraphWrappers:** Infrastructure layer for building LangGraph agents with reusable components (nodes, edges, tools, structured outputs). Demonstrates systems-level thinking applied to agent development frameworks. [GitHub](#).
- **StocksMarketRecommender:** Multi-agent workflow with Yahoo Finance MCP integration. Orchestrates multiple LangGraph flows using LangGraphWrappers library. [GitHub](#).
- **SoccerSmartBet (ongoing):** Real-time betting analysis system comparing AI vs. human strategies. Demonstrates high level architecture of complex independent agentic workflows using LangGraph infrastructure. [GitHub](#).
- Completed **AI Engineer Agentic Track: The Complete Agent & MCP Course** (Udemy, 2025) - comprehensive training in AI agents, multi-agent systems, MCP protocol, prompt engineering, and AI systems design.
- Published technical series on database implementation and C++ techniques on LinkedIn, republished by [EuropeClouds](#) (2020-2021). Created [C++ TMP Tutorial](#) referenced in articles.